

Progress report (Phase 2_6th Semester) of the Final Year Project titled

Design and Implementation of an Automated Antenna Radiation Pattern Measurement System using Embedded Control and Image Processing

Submitted for partial

Fulfillment of the requirements for the degree of

Bachelor of Technology

in

Electronics and Communication Engineering[ECE-3]

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Project Title: Design and Implementation of an Automated Antenna Radiation Pattern Measurement System using Embedded Control and Image Processing

Objective:

- **Automated Rotation Control:** To make a web UI to control antenna position using a **NEMA 17 stepper motor** and **A4988 motor driver** for 10 ° incremental rotation.
- **Non-Invasive Data Acquisition And Extract And Manipulation:** To capture current meter readings via **ESP 32S CAM** and **ESP 32 CAM MB** , ensuring zero electrical interference with the antenna and extract numerical measurement data from captured images using computer vision and image processing techniques then store and calculate the relative radiated power in decibels (dB) from stored data .
- **Automated Pattern Plotting:** To design a motor-driven CNC machine plotter that physically plots the final radiation pattern on paper, eliminating manual graphing errors.

Scope of Work and Identification of Domain of Project:

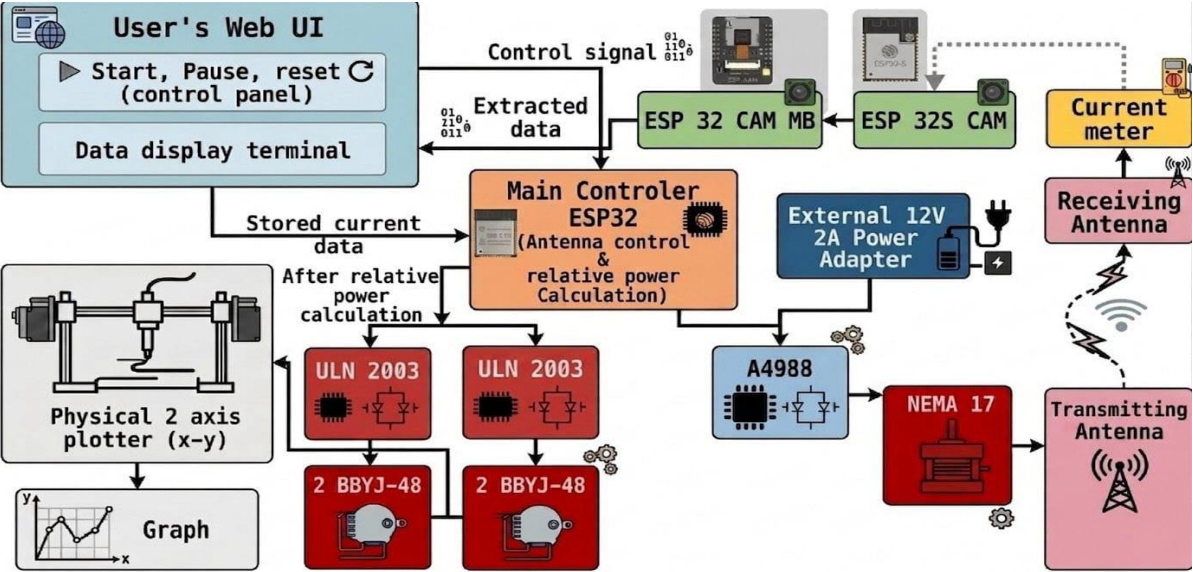
1. **Antenna Rotation Control:** Program a ESP 32 and make a web UI to control a NEMA 17 stepper motor that rotates the transmitter antenna in precise 10-degree increments through a full 360-degree rotation (36 positions). At each position, the system pauses for exactly 30 seconds to allow current stabilization before image capture.
2. **Computer Vision Data Acquisition & Processing:** Integrate a camera module with the embedded system using ESP 32S CAM and ESP 32 CAM MB. The system will:
 - Capture an image of the current meter display at each 10-degree position after the 30-second stabilization period.
 - Process the image using OCR algorithms to extract the numerical current value and display it on the web UI.
 - Then the extracted data will be stored and then from stored data we will Calculate the relative power in dB (in ESP 32) using the formula:
$$\text{Power (dB)} = 20 \times \log(\text{Current Reading}/N), \text{ where } N = \text{the maximum current reading obtained from all 36 positions}$$
 - Store the data triplet (Angle, Current Reading, Calculated dB Power) in microcontroller memory.
2. **Physical Graph Plotting:**
 - Activate a 2-axis pen plotter.
 - Use stored angle and calculated dB power data to control plotter motors.
 - Draw a complete rectangular coordinate graph (Angle vs. dB Power) on paper.
 - Ensure the graph clearly shows the normalized radiation pattern relative to maximum signal strength.

Thus, the project lies at the intersection of:

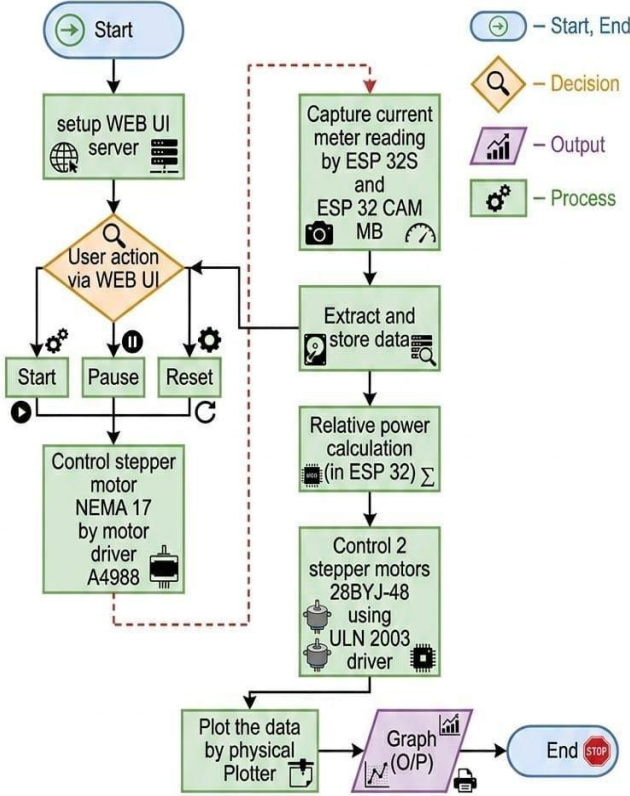
- ✓ Embedded automation
- ✓ Computer vision and image processing
- ✓ RF measurement system

Block Diagram:

Embedded Automated Antenna Radiation Pattern Measurement System



Workflow:



Initial Conditions:

- The camera is positioned with proper lighting and focus to capture clear images of the meter display.
- The 30-second stabilization period is sufficient for current reading to become stable at each position.

Target Customer:

- IoT Wireless Module and Sensor Developers
- Automated RF labs
- Educational Training Kit Manufacturers

Technology Used:

- Automation and Control Technology
- Computer Vision and Image Processing
- Plotting Mechanism

Component Description:

1. Central Processing Unit
 - ESP 32 (512MB RAM) - Main controller
2. Mechanical Rotation System
 - NEMA 17 Stepper Motor - Rotates antenna
 - A4988 Motor Driver - Controls antenna rotation motor
3. Computer Vision and Image Acquisition
 - ESP 32S CAM - Captures meter images
 - ESP 32 CAM MB - Connector
4. Physical Plotting System
 - 28BYJ-48 Stepper Motor(X-axis) - Horizontal movement
 - ULN2003 Driver Board (X-axis) - X-axis motor control
 - 28BYJ-48 Stepper Motor (Y-axis) - Vertical movement
 - ULN2003 Driver Board (Y-axis) - Y-axis motor control
 - Linear Rails (2 units) - 300mm aluminium guides
 - Timing Belts - GT2 timing belts with pulleys
 - SG90 Micro Servo - Pen lift mechanism
 - Pen Holder - Spring-loaded mechanism
 - Drawing Surface - A4 paper on rigid backing
 - Frame Structure - Acrylic/aluminium frame
5. Power Supply System
 - 12V 4A Power Adapter - USB-C connector for ESP 32

Feasibility Analysis :

- Technical Feasibility
 - NEMA 17 Stepper motors ensure precise angular control.
 - ESP 32S CAM enables non-contact data capture.
 - CNC machine plotter plots the graph physically.
- Operational Feasibility
 - Fully automated operation via single switch.
 - Minimal human supervision required.
 - Does not require complex RF interfacing.
- Economic Feasibility
 - Uses low-cost and easily available components.
 - Avoids expensive antenna measurement instruments.

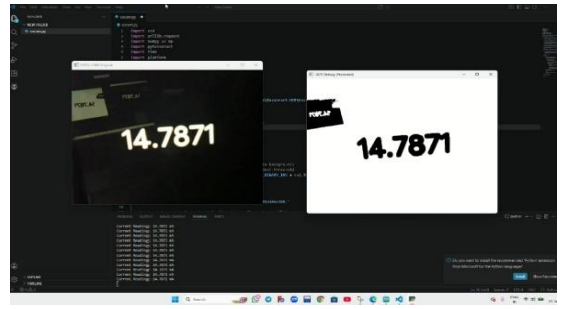
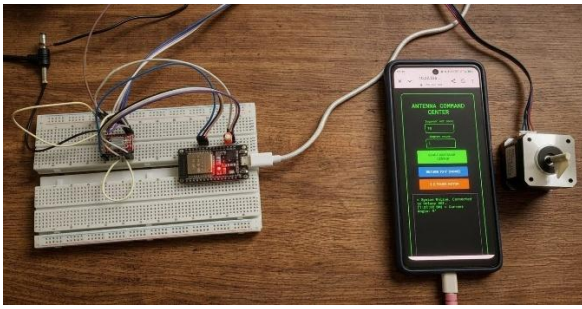
Literature Survey:

- R. C. Johnson
In his work on antenna measurement systems, Johnson highlighted the limitations of manual radiation pattern measurement methods and emphasized the need for automated positioning systems to improve angular accuracy.
Reference Book : Antenna Engineering Handbook – R. C. Johnson
- Constantine A. Balanis
In Antenna Theory: Analysis and Design, Balanis discussed radiation pattern characterization techniques and measurement setups. He noted that traditional measurements require mechanical precision and suffer from human reading errors.
Reference Book : Antenna Theory: Analysis and Design – Constantine A. Balanis
- S. C. Orfanidis
Orfanidis introduced digital methods in RF testing and discussed how measurement automation can enhance repeatability and reduce observational inaccuracies in electromagnetic experiments.
Reference Book : Electromagnetic Waves and Antennas – S. C. Orfanidis
- J. Fraden
In sensor-based measurement systems research, Fraden explored non-contact sensing techniques, suggesting optical methods as reliable alternatives when direct electrical interfacing is not feasible.
Reference Book : Handbook of Modern Sensors – Jacob Fraden
- Zhang et al.
Proposed vision-based instrument reading systems using OCR to eliminate manual monitoring. Their research showed improved flexibility when image processing replaces wired data acquisition.
Link : <https://ieeexplore.ieee.org/document/8461184>
- Y. Yalçin, S. Kurtulan
Developed motor-controlled positioning systems for antenna scanning that improved angular precision but lacked automated data extraction capability.
Link : <https://doi.org/10.1109/MAP.2009.5162072>

Gap Identification:

- ✓ **Lack of Fully Integrated Automation:** Existing systems automate either rotation OR measurement not both together.
- ✓ **Dependency on Electrical Interface:** Most systems require wired connection to meters, making them less flexible.
- ✓ **Absence of Self-contained Embedded Operation:** Many setups rely on PC-based processing.
- ✓ **No Direct Physical Visualization:** Graphs are digitally plotted but not physically drawn.
- ✓ **Human Intervention Still Required:** In positioning, reading, or plotting stage.

Prototype Development:



The prototype of our project has been successfully developed and tested. It consists of an ESP32 microcontroller, a stepper motor with driver module, and a mobile/web-based control interface for real-time operation.

The ESP32 hosts a local web server that allows the user to control the antenna movement through a smartphone interface. Commands such as start rotation, pause motor, and return to home position have been implemented and tested successfully.

A NEMA 17 stepper motor is used to provide precise angular movement of the antenna system. The motor rotation is controlled through programmed step sequences, ensuring accurate positioning and repeatable motion.

In addition, an ESP32-CAM and OCR-based monitoring system has been integrated to capture the display readings and extract numerical values automatically using image processing techniques. The OCR module successfully detects and displays the measured values in real time, which improves monitoring accuracy without electrical connection and reduces manual observation.

The prototype has demonstrated successful communication between the hardware and software modules, proving the feasibility of the proposed system design.

Quality of Documentation:

- It includes a clear project title, objectives, scope, and workflow.
- We have provided block diagrams and system workflow diagrams for better understanding.
- The documentation covers:
 - problem statement
 - technology used
 - component description
 - feasibility analysis
 - literature survey
 - gap identification
- Each subsystem such as:
 - motor control
 - image processing
 - plotter mechanism
 - power system is properly used
- The report also includes technical justification and references from standard books and IEEE papers.
- This improves the readability, reproducibility, and technical credibility of the project.